

Mathew Jacob

PHD STUDENT · UNIVERSITY OF WASHINGTON

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Education

University of Washington

PHD COMPUTER SCIENCE

- Advisors: Stephanie Wang, Luis Ceze

Seattle, WA

2025 - present

University of Illinois Urbana-Champaign

BS COMPUTER SCIENCE

- Research Advisor: Laxmikant Kale

Champaign, IL

2021 - 2025

Professional Experience

University of Washington

GRADUATE RESEARCHER

- Exploring how to unify training and inference frameworks to ease training and deployment of ML models, specifically focusing on the reinforcement-learning workloads.
- Helping to develop a new scheduling language to make it easier to try new pipeline parallelism training schedules. This simpler language will not only ease manual testing of different pipeline parallelism schedules, but also serve as a new interface for ML models to automatically find new parallelism schedules that previously were hard to find.

Seattle, WA

2025-present

Parallel Programming Lab, UIUC

RESEARCH ASSISTANT

- Worked on improving parallel read performance in Charm++.
- Explored supporting streaming applications in Charm++ and utilizing the HPC specific features to improve other workloads that Charm++ traditionally isn't used for.

Urbana, IL

2022-2025

Databricks

RESEARCH SCIENTIST INTERN

- Advised by Matei Zaharia, Michael Carbin, Omar Khattab, and Andrew Drozdov
- Explored reranker quality characteristics to better understand their characteristics and how to effectively deploy them in production.
- Constructed various pareto curves for RAG systems that are used to inform products.
- Formulated discoveries in a paper about properties of rerankers in RAG systems.
- Work integrated into various products: see the Databricks reranker: <https://www.databricks.com/blog/reranking-mosaic-ai-vector-search-faster-smarter-retrieval-rag-agents> and the Databricks Agent Bricks Knowledge Assistant: <https://docs.databricks.com/aws/en/generative-ai/agent-bricks/knowledge-assistant>.

San Francisco, CA

Summer 2024

Databricks

SOFTWARE ENGINEERING INTERN

- Built the DeepspeedTorchDistributor, enabling Deepspeed training/inference on Spark clusters. Featured in blog post: <https://community.databricks.com/t5/technical-blog/introducing-the-deepspeed-distributor-on-databricks/ba-p/59641> here
- Contributed to HF datasets by making the Spark dataframe reads faster via prefetching.

San Francisco, CA

Summer 2023

Papers and Manuscripts

Melissa Z. Pan, Negar Arabzadeh, **Mathew Jacob**, Fiodar Kazhamiaka, Esha Choukse, Matei Zaharia. 2025. Towards Automatically Optimizing Retrieval Augmented AI Systems. To appear in *ML For Systems Workshop at NeurIPS 2025*

Mathew Jacob, Erik Lindgren, Matei Zaharia, Michael Carbin, Omar Khattab, and Andrew Drozdov. Drowning in Documents: Consequences of Scaling Reranker Inference. In *ReNeuIR 2025 Workshop at SIGIR 2025*. arXiv preprint arXiv:2411.11767. <https://arxiv.org/abs/2411.11767>.

Mathew Jacob, Maya Taylor, and Laxmikant Kale. 2024. "CkIO: Parallel File Input for Over-Decomposed Task-Based Systems." arXiv preprint arXiv:2411.18593, November 27, 2024.

Mathew Jacob. 2021. Eir: A Python Package for Epidemic Simulation. *Journal of Open Source Software*, 6(62), 3247. DOI: 10.21105/joss.03247.

Open Source Contributions

- 2023

Apache Spark, Built DeepspeedTorchDistributor and maintained contributed to ML capabilities of Spark.

GitHub
- 2023

Huggingface Datasets, Improved converting spark dataframes to huggingface datasets

GitHub

Awards, Fellowships, & Grants

- 2025

First-Year Graduate Research Endowment, Jonathan Hsieh and Kathleen Ting Endowed Fund for Graduate Students in Computer Science & Engineering

\$ 3,000

Presentations

** presenting author; + mentored undergraduate*

CONTRIBUTED PRESENTATIONS

- Mathew Jacob**, Erik Lindgren, Matei Zaharia, Michael Carbin, Omar Khattab, Andrew Drozdov*. 2025. Drowning in Documents: Consequences of Scaling Reranker Inference. *ReNeuIR at SIGIR 2025, Padova, Italy*
- Mathew Jacob**, Maya Taylor*, Laxmikant Kale. CkIO: Parallel File Input for Over-Decomposed Task-Based Systems. 2024 *JLESC Workshop*